

Imperfect Competition: Oligopoly & Monopolistic Competition

C211: Global Economics for Managers
Western Governors University



The History of Monopolistic Competition

- Perfect competition
 - Early Industrial Revolution Capitalism
 - Firms are owner operated, producing basic commodities
 - Competition on price only
- From this brands begin to develop
 - Easy way to differentiate between quality of goods
 - Example: “I know these are good oats because they are Quaker”



Early Ads

- Focused on quality of product
- Focused on trustworthiness
- Created brand loyalty in customers
- Gave market power to trusted/popular brands



[Click Image for source reference](#)

Monopolistic Competition

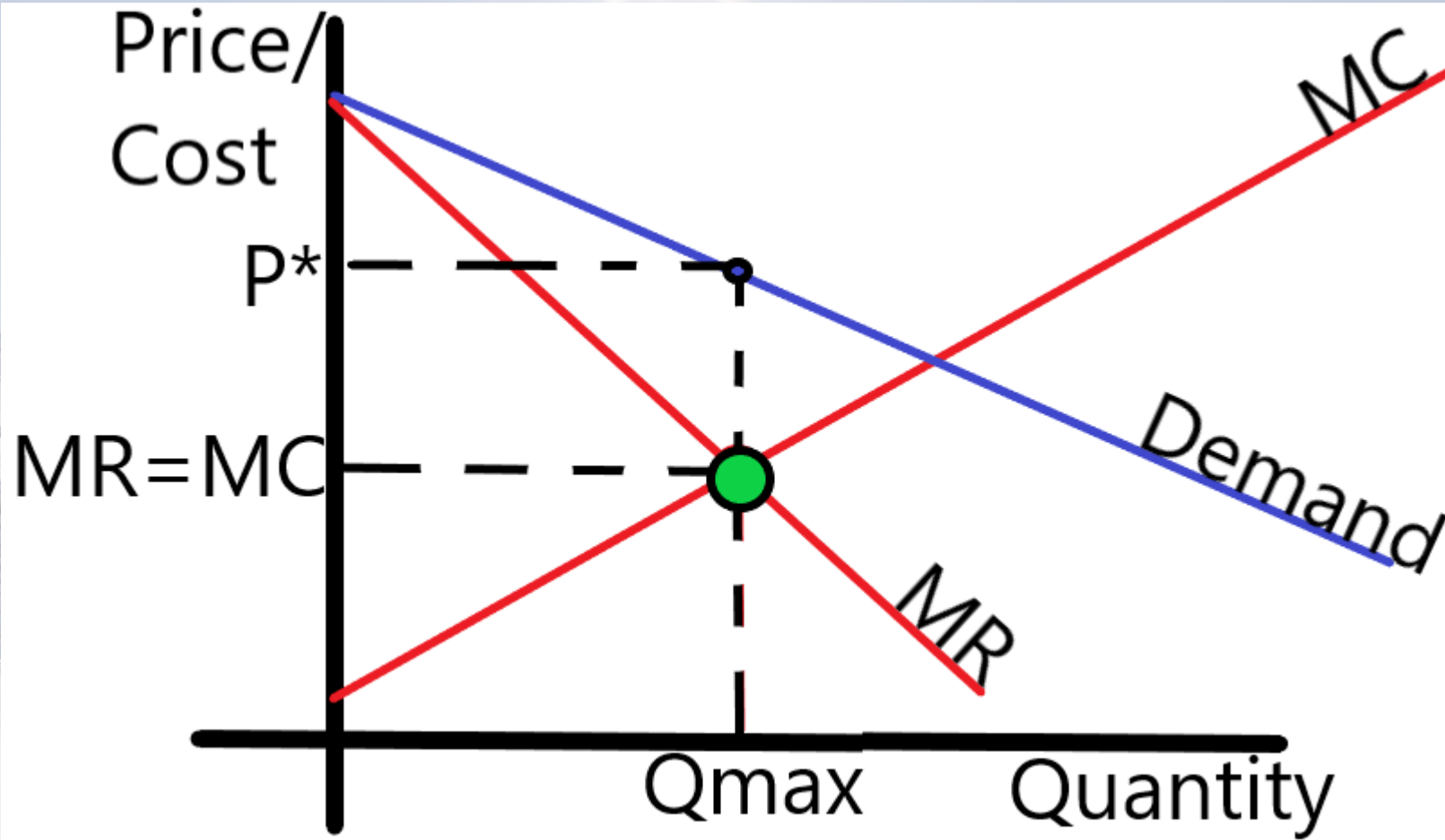
- Developed by Joan Robinson in the early 1900s
- Captures the power of brand loyalty and the ability to affect market price
- People will pay a premium for a brand they prefer
- It combines Perfect Competition and Monopoly features

Monopolistic Competition Basics



- **Differentiated good**
- **No barriers to entry**
 - -Cheap and easy to enter
- **Can affect market price**
 - -Individual firms changing quantity changes price
- **$P > MR$**
- **$P > MC$ (at profit max)**
- **Examples:**
 - -Fast Food, Athletic Gear

Monopolistic Graph



- Terms

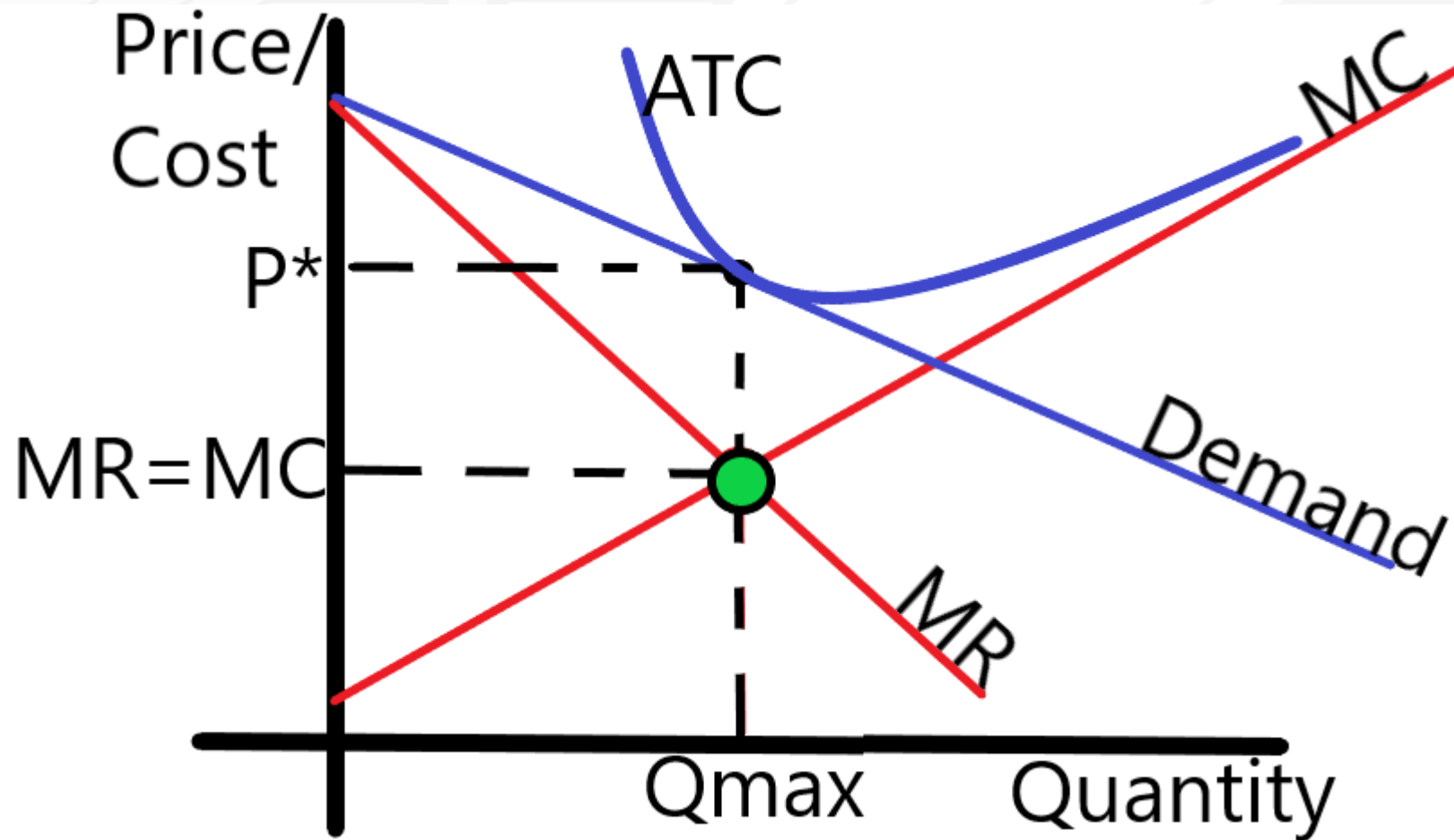
MC → Marginal Cost

MR → Marginal Revenue

- **Qmax** → profit maximizing Quantity

- **P*** = Price charged by firm

Monopolistic Long-Run Graph



- ATC → Average Total Cost

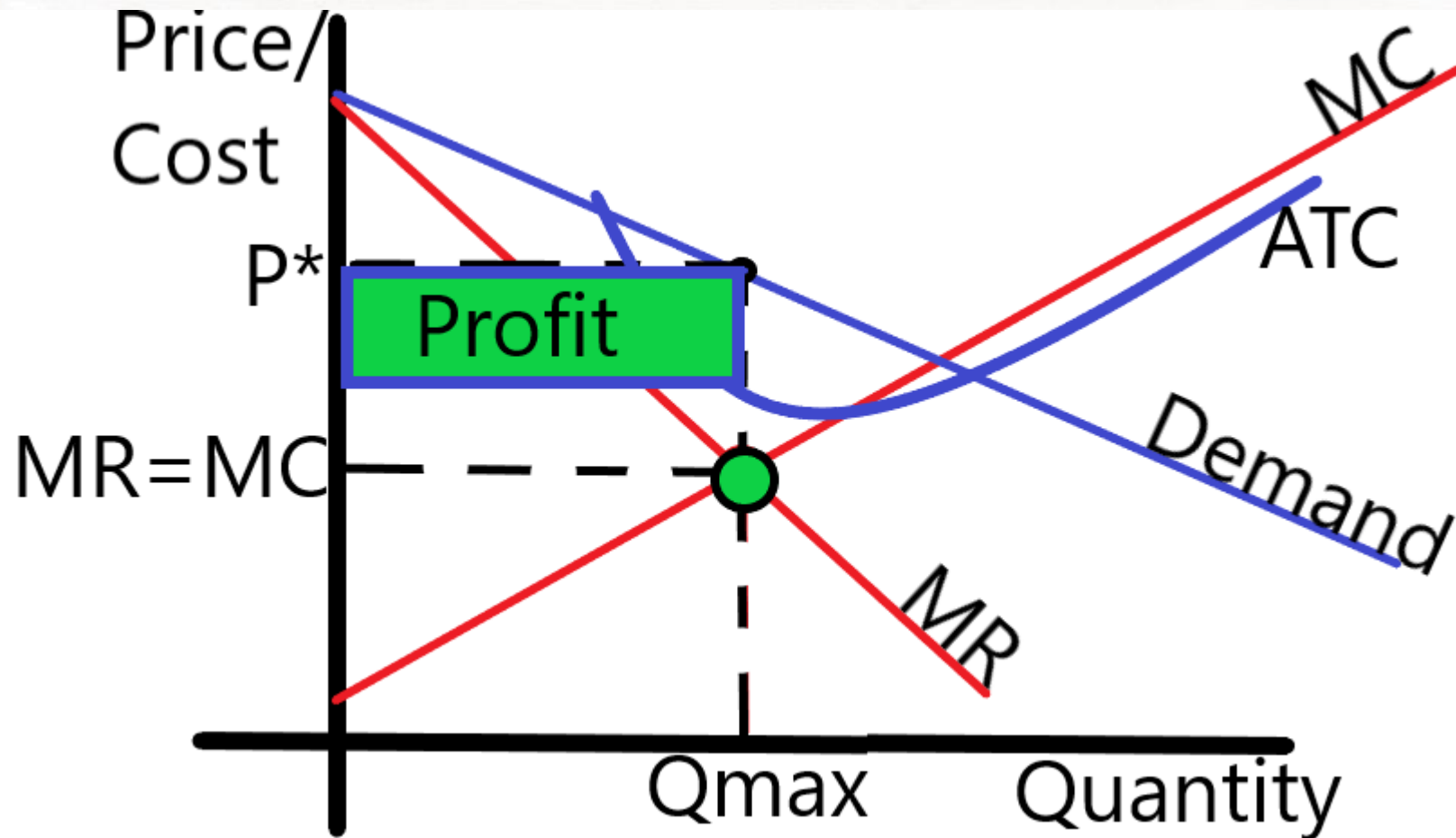
- $P^*=ATC$

- No profit being earned

- Competition moving into and out of market eliminates profits

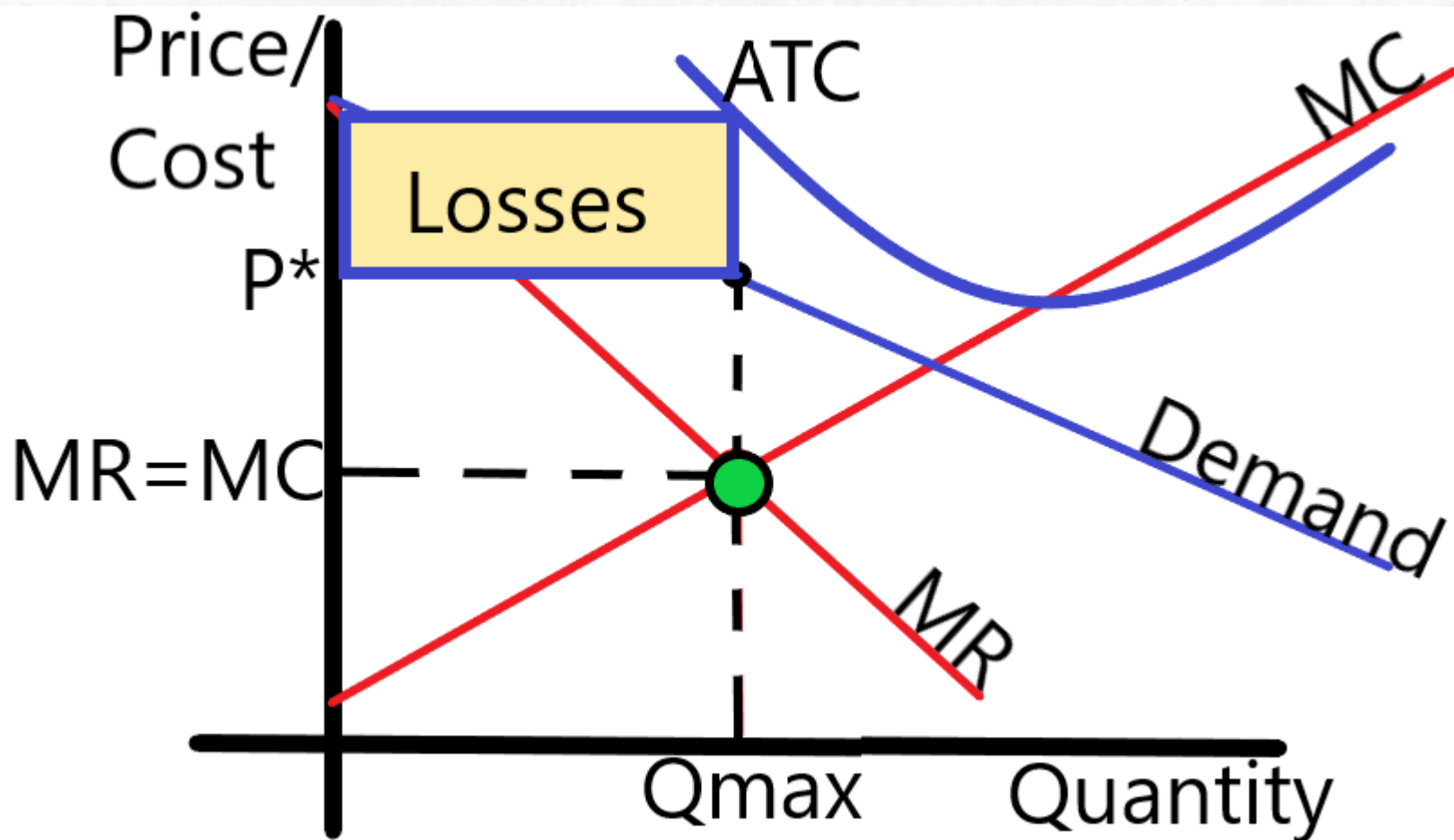
- See CH16-2b

Short-Run (Possibility 1)



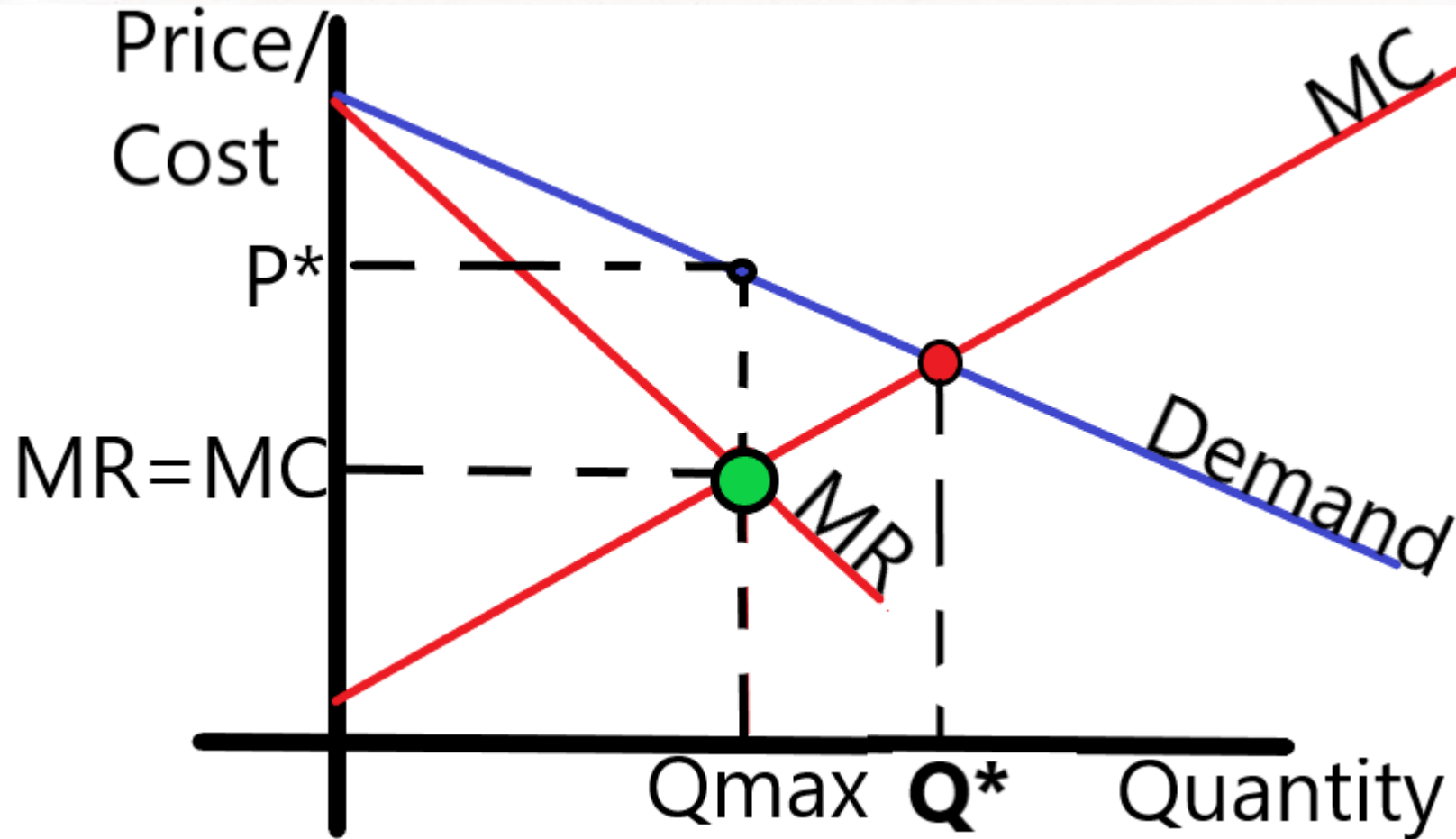
- $P^* > ATC$
 - Price received more than covers costs
- This will encourage new competition
 - As new firms enter the profit disappears
- The Demand curve will move to left
 - as new options become available to customers
- See Ch16-2b

Short-Run (Possibility 2)



- $P^* < ATC$
 - Price received does not covers costs
- This will cause some firms to Shut-Down and some to Exit
- The Demand curve will move to right
 - as fewer options become available to customers
- See Ch16-2b

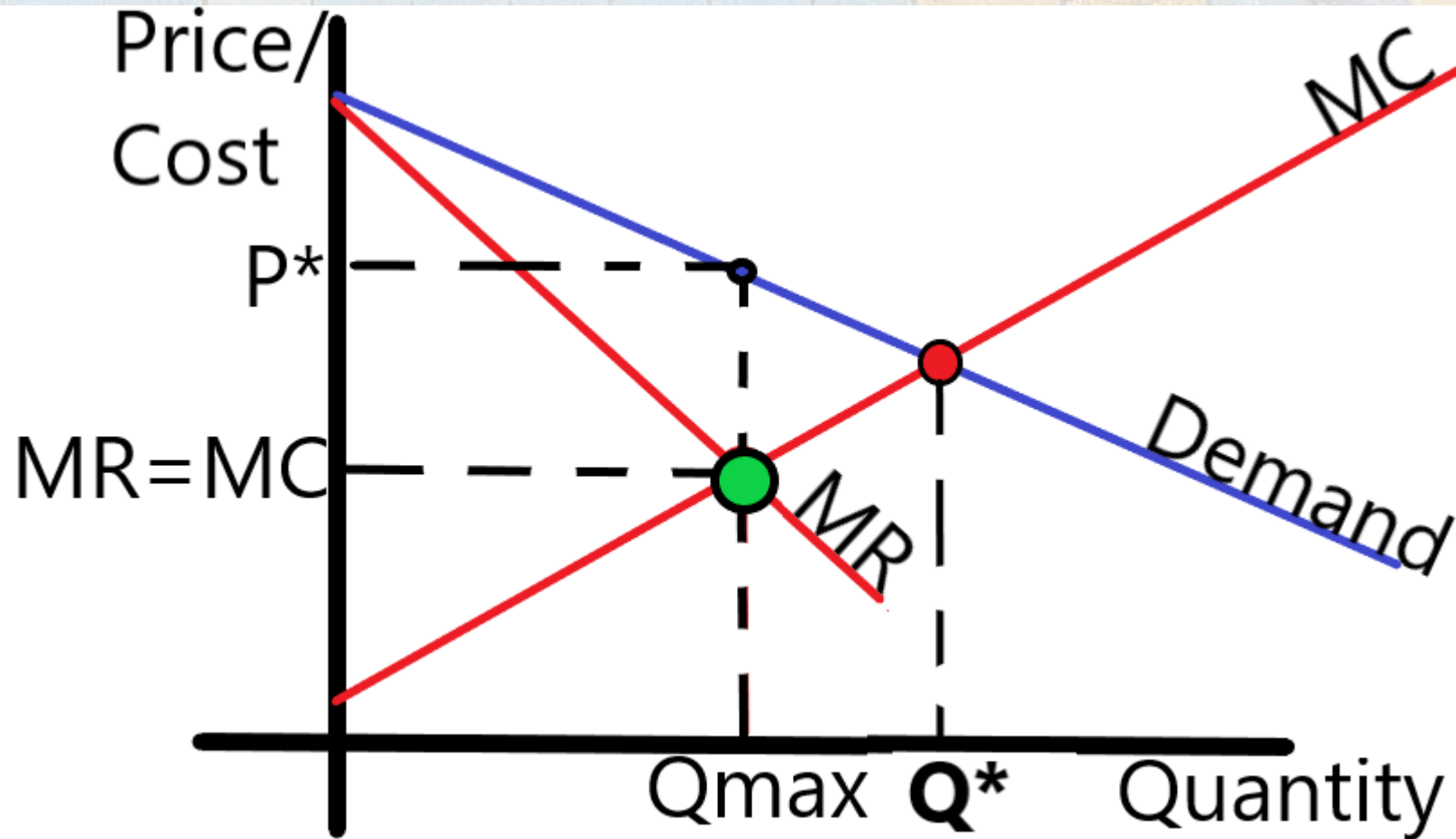
Inefficient



- Q^* =Efficient Quantity
- Monopolistic firms can affect Price
 - They reduce production
 - To get a higher Price
- Produce less than efficient quantity

See Ch16-2c

Excess Capacity & The Markup



- Q^* =Efficient Quantity

- **Excess Capacity**

- The difference between Q_{max} and Q^* is called
- Represents the fact they are artificially reducing quantity

- The Markup

- The difference between P^* and MC

See Ch16-2c

The History of Oligopoly

- Developed from ‘Game Theory’
- Developed by John Nash in the 1950s
- Tried to explain how some markets have high but not impossible barriers
 - And yet can still be considered competitive
- Used the story of fictional prisoners

The Prisoner's Dilemma

Two Prisoners

Separate interrogation
rooms

Can't talk with each other

Given a choice; Confess, or
remain silent

The Prisoner's Dilemma

Prisoner A

Prisoner B

	Confess	Silent
Confess	Both Confess	A Remains Silent & B Confesses
Silent	B Remains Silent & A Confesses	Both Remain Silent

The Prisoner's Dilemma

Prisoner A

Prisoner B

	Confess	Silent
Confess	Both Get 5 Years	A gets 20 years & B goes Free
Silent	B Gets 20 Years & A Goes Free	Both Get 1 Year

The Prisoner's Dilemma

		Prisoner A	
		Confess	Silent
Prisoner B	Confess	5 Years For A 5 Years For B	20 Years For A N0 Years For B
	Silent	N0 Years For A 20 Years For B	1 Year For A 1 Year For B

The Prisoner's Dilemma

Prisoner A			
Prisoner B	Confess	Silent	
	Confess	5 Years For A	20 Years For A
If Prisoner B Confesses			
If A Confesses they get 5 years If A remains Silent they get 20 years			
So A decides Confess is the best choice			

The Prisoner's Dilemma

Prisoner A			
Prisoner B		Confess	Silent
	If Prisoner B Remains Silent		
	If A Confesses they get N0 years If A remains Silent they get 1 year		
So A decides Confess is the best choice			
Silent		N0 Years For A	1 Year For A

The Prisoner's Dilemma

		Prisoner A	
		Confess	Silent
Prisoner B	Confess	5 Years For A 5 Years For B	20 Years For A NO Years For B
	Silent	NO Years For A 20 Years For B	1 Year For A 1 Year For B

The Prisoner's Dilemma

Prisoner A

Prisoner B

	Confess
Confess	5 Years For B
Silent	20 Years For B

If Prisoner A Confesses

If B Confesses they get 5 years

If B remains Silent they get 20 years

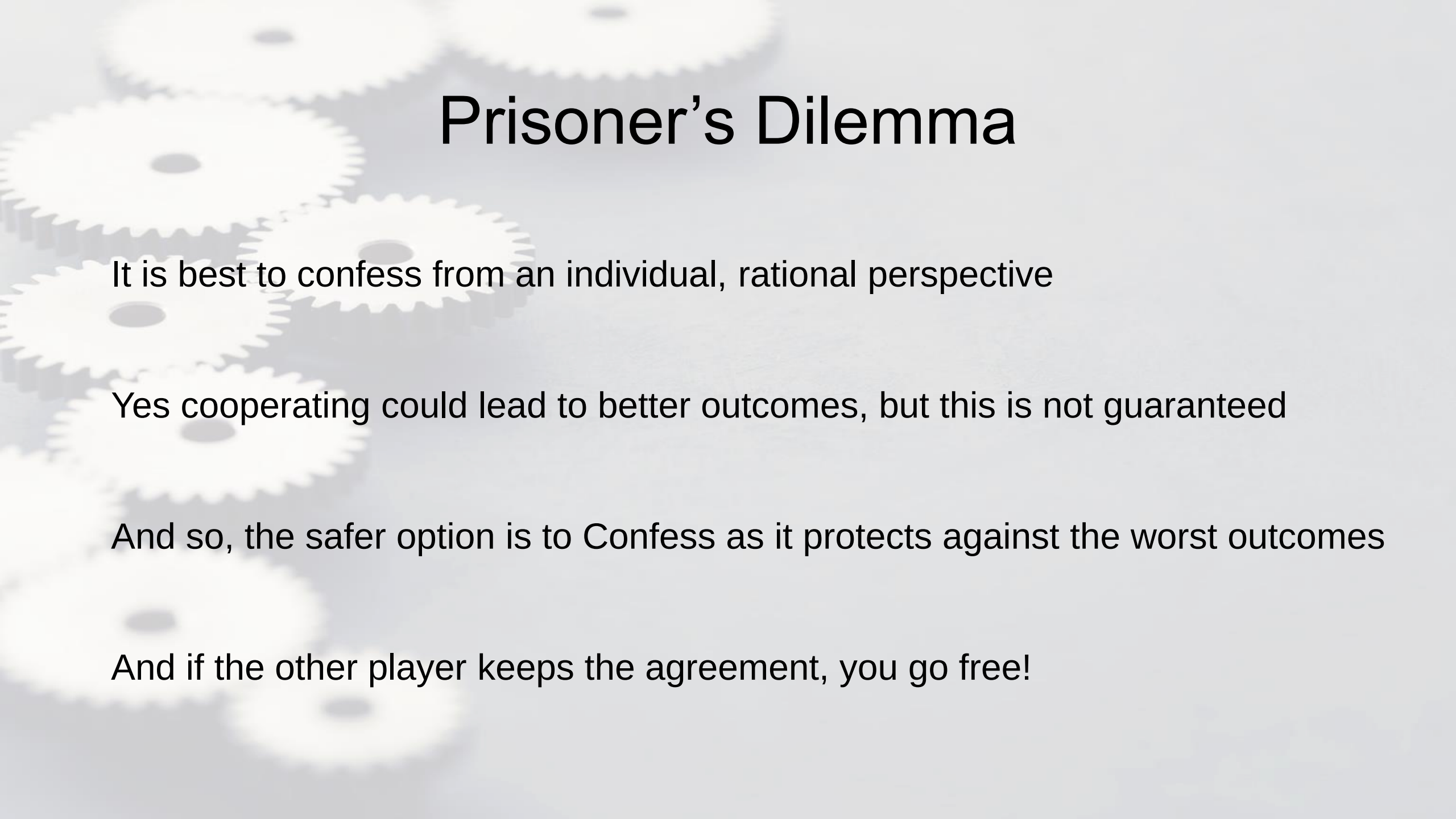
So B decides Confess is the best choice

The Prisoner's Dilemma

Prisoner A

Prisoner B

		Prisoner A	
		Confess	Silent
Prisoner B	Confess	If Prisoner A Remains Silent If B Confesses they get N0 years If B remains Silent they get 20 years	N0 Years For B
	Silent	So B decides Confess is the best choice	1 Year For B



Prisoner's Dilemma

It is best to confess from an individual, rational perspective

Yes cooperating could lead to better outcomes, but this is not guaranteed

And so, the safer option is to Confess as it protects against the worst outcomes

And if the other player keeps the agreement, you go free!

The Prisoner's Dilemma: Duopolies

Choosing Low Quantity is Monopoly-like.

Low Quantity leads to a higher price, less efficient markets but higher profits.

Business B

Business A

	High Q	Low Q
High Q	Low Profit Low Profit	Very High Profit Very Low Profit
Low Q	Very High Profit Very Low Profit	High Profit High Profit

Choosing High Quantity is acting more competitive.

High Quantity leads to a lower price, more efficient markets and is better for consumers but lower profits.

The Prisoner's Dilemma: Duopolies

Business A

		Business A	
		High Q	Low Q
Business B	High Q	Low Profit	Very Low Profit
	Low Q	Low Profit	Low Profit

Business B Chooses High:

If A chooses Low they get Very Low Profit
If A chooses High they get Low profit

So A decides High Q is best

The Prisoner's Dilemma: Duopolies

Business A

Business B

	High Q	Low Q
Low Q	Very High Profit	High Profit

Business B Chooses Low:

If A chooses High they get Very High Profit
If A chooses Low they get High profit

So A decides High Q is best

The Prisoner's Dilemma: Duopolies

Business A

		Business A	
		High Q	Low Q
Business B	High Q	Low Profit	If Business A Chooses High: If B Chooses High they get Low Profits If B chooses Low they Very Low Profits So B decides High is their best choice
	Low Q	Very Low Profit	

The Prisoner's Dilemma: Duopolies

Business A

		High Q	Low Q
Business B	High Q	If Business A chooses Low: If B chooses High they get Very High Profits If B chooses Low they get High Profits So B decides High is the best choice	Very High Profit
	Low Q		High Profit

The Prisoner's Dilemma: Duopolies

Business A

Rational players would choose High Q REGARDLESS of what the other player does or whichever they choose

Business B

	High Q	Low Q
High Q	Low Profit Low Profit	Very High Profit Very Low Profit
Low Q	Very Low Profit Very High Profit	High Profit High Profit

The Nash Equilibrium is High Q, High Q

It is the best option GIVEN what the other player does and so cannot be improved

Duopolies: Two Ways to maintain Collusion

1) Repeated Interactions

Uses
“Signaling”

If the players can repeat the interaction, they will figure out how to successfully collude

Business A

		Business A	
		High Q	Low Q
Business B	High Q	Low Profit Low Profit	Very High Profit Very Low Profit
	Low Q	Very Low Profit Very High Profit	High Profit High Profit

2) Capacity to Punish

Also called
“Mutual Forbearance”

If one player can hurt the other's profits in another market, they can force collusion

Conclusion

- Monopolistic Competition

- Low barriers to entry (lots of competition, many firms)
- $P > MR$ and $P > MC$ (at profit max)
- Can earn profits in short-run but not in long-run
- Differentiated product, in other words BRANDS!
- Produces less than the efficient scale

- Oligopoly

- High but not impossible barriers (only 2-5 firms)
- Assumed inefficient, $P > MR$ etc.
- Nash Equilibrium is sub-optimal for players
 - But rational
- Collusion is hard to maintain because of self-interest and competition